



Digital Forensic Internship-2026

Evidence Acquisition & Imaging – Legal & Admissibility

Forensic Investigation Report

Task: Week2 – Task2

Task Title: Evidence Acquisition & Imaging – Legal & Admissibility

Task Overview: This task focuses on understanding how digital evidence is collected and preserved in a legally acceptable manner under Pakistani law. It examines the process of evidence acquisition, forensic imaging, hashing, and chain of custody, and maps these technical procedures to the requirements of the Prevention of Electronic Crimes Act (PECA) 2016 and the Qanun-e-Shahadat Order (QSO). The objective is to identify how each step helps maintain the integrity, authenticity, and admissibility of digital evidence in court, while also highlighting the major risks that could make evidence inadmissible if proper procedures are not followed.

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1. Introduction / Objective

Digital evidence such as forensic disk images, hash values, and system logs plays an increasingly central role in cybercrime investigations in Pakistan. However, the technical soundness of an imaging or hashing process does not, by itself, guarantee that the resulting evidence will be accepted in court. Admissibility depends on whether the process complies with the legal requirements set out under the Qanun-e-Shahadat Order (QSO), 1984 and the Prevention of Electronic Crimes Act (PECA), 2016.

This report examines the standard technical steps involved in evidence acquisition and imaging, and maps each step to the corresponding legal requirement it is intended to satisfy. It further identifies the key points at which improper handling can compromise the admissibility of otherwise valid evidence, drawing on relevant statutory provisions and reported case law. The objective is to demonstrate that legal admissibility is not a separate afterthought to forensic work, but a direct outcome of how correctly the acquisition process is executed and documented.

2. Technical Process Overview

When digital evidence such as a computer, phone, or storage device is seized during an investigation, it goes through a controlled process before it can be analyzed or presented in court. This process is designed to make sure the evidence stays exactly the same from the moment it's collected to the moment it's shown to a judge.

The typical steps are:

- **Seizure** — The device is taken into official custody, and its condition at the time of seizure (powered on/off, physical state, location) is documented.
- **Write-blocking** — Before anything is copied, a write-blocker is used to make sure the original device cannot be accidentally changed, even by something as small as plugging it into a computer.
- **Imaging** — An exact, bit-by-bit copy of the entire device is created. This copy includes everything — even deleted files and hidden system data — not just the visible files.
- **Hashing** — A unique digital fingerprint (using algorithms like SHA-256) is generated for both the original device and the copy. If the fingerprints match, it proves the copy is identical to the original.
- **Analysis** — All examination work is done only on the copy. The original is sealed and stored untouched, so it remains available for verification later if needed.
- **Documentation** — Every action taken, and every person who handles the evidence, is logged in writing from the moment of seizure onward.

This process exists for one reason: to make sure that what's examined and shown in court is provably the same as what was originally found — nothing added, nothing missing, nothing altered.

4. Legal Framework Overview

4.1 Qanun-e-Shahadat Order (QSO), 1984 — Admissibility of Electronic Evidence

QSO 1984 is Pakistan's general law of evidence. It was originally written before digital evidence existed, so specific provisions were later introduced to give electronic evidence legal recognition:

- *Article 164* permits courts to admit evidence produced through modern devices or techniques (e.g., forensic images, CCTV, digital records), at the court's discretion.
- *Article 73* classifies electronic documents as primary evidence rather than secondary/copied evidence.
- *Article 78-A* addresses the certification/attestation requirements for electronic documents to be treated as genuine.

Together, these articles establish that digital forensic evidence is legally recognized in Pakistani courts, provided its authenticity can be demonstrated.

4.2 Prevention of Electronic Crimes Act (PECA), 2016 — Investigation and Seizure Procedure

While QSO governs *what* can be admitted, PECA 2016 governs *how* electronic evidence must be investigated, seized, and handled. Key aspects relevant to this task include:

- Authorization requirements for seizing devices or data during an investigation.
- Procedures for handling seized data/information systems securely post-seizure.
- A prescribed **Chain of Custody form (Schedule VI)** under the PECA Rules, used to formally document every transfer of evidence from seizure through to court presentation.
- Requirements for forensic lab handling of digital evidence.

PECA compliance is what gives the technical imaging/hashing process legal weight — it converts routine forensic steps into a documented, legally defensible procedure

5. Case Law

The following cases illustrate how Pakistani courts have applied the admissibility standards discussed above:

- **The State v Ahmed Omer Sheikh** — A recording was presented without disclosing its source or forensically confirming it had not been altered. The court held that such evidence carried no evidentiary value, reinforcing that authenticity must be demonstrable, not assumed.

- **Mian Khalid Pervaiz v The State (2021 SCMR 522)** — The Supreme Court confirmed that digital evidence is admissible under QSO's Article 164 and 78-A framework, provided the correct procedure for producing and proving such evidence is followed.
- **Malik Khalid Pervez v State** — Call data records (CDR) were submitted without supporting transcripts or recordings. The court held that raw call logs alone were insufficient and could not be relied upon as evidence.

7. Mapping Technical Steps to Legal Requirements

Technical Step	Legal Requirement	What Proves Compliance	Risk If Skipped
Seizure & Documentation	PECA 2016 – authorized seizure procedure	Seizure memo, officer's authorization, device condition noted at time of seizure	Evidence can be challenged as illegally obtained; may be excluded entirely regardless of content
Write-blocking	QSO Art. 164 & 78-A – proof evidence wasn't altered	Write-blocker log/report showing original device was never written to	Defense can argue the original was modified during handling, destroying trust in everything after it
Forensic Imaging	QSO Art. 73 – electronic copy treated as primary evidence	Imaging tool report (e.g., FTK Imager log) confirming a full, unmodified bit-by-bit copy	If imaging isn't complete/verified, the copy may be treated as unreliable secondary evidence, not primary
Hashing (before & after)	QSO Art. 78-A – authenticity/integrity certification	Matching hash values (original vs. image) recorded and preserved	No hash match = no proof the evidence is untampered; this is the single most common reason digital evidence gets rejected
Analysis (on copy only)	QSO Art. 164 – reliability of modern-device evidence	Analysis log showing original was never touched, only the verified image was examined	If original was altered during analysis, entire evidence chain becomes unreliable
Chain of Custody Documentation	PECA 2016 – official Chain of Custody form (Schedule VI)	Signed custody log showing every handler, date, time, and reason for access	Any gap or missing signature lets the defense argue the evidence could have been tampered with during that gap

Court Presentation	QSO Art. 164 & 73 – final admissibility decision	Testimony of the examiner + complete documentation trail presented together	Without a complete, defensible trail, even technically perfect evidence can be ruled inadmissible
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8. Top Admissibility Risks

Based on the technical-legal mapping and supporting case law, the following are the most significant risks to the admissibility of digital evidence:

1. **Broken Chain of Custody** — Any gap or missing documentation in the handling timeline allows the defense to argue possible tampering, even without proof it occurred.
2. **Missing or Mismatched Hash Verification** — Without matching hash values between the original and the image, there is no reliable proof the evidence remains unaltered, directly undermining authenticity requirements under QSO Article 78-A.
3. **Analysis Performed on the Original Device** — Examining the original instead of a verified copy risks altering the evidence itself and invites challenges to its integrity.
4. **Undisclosed Source or Origin of Evidence** — Evidence presented without a clear, provable source has been rejected by courts outright, regardless of its apparent relevance.
5. **Incomplete Evidence Submission** — Submitting partial evidence (e.g., logs without supporting recordings or transcripts) has been held insufficient for reliance, even when the partial evidence itself is genuine.

9. Conclusion

This analysis shows that the admissibility of digital evidence in Pakistani courts depends less on the sophistication of the forensic tools used, and more on whether each technical step can be proven to have been done correctly and documented without gaps. QSO 1984 establishes that digital evidence is legally recognized, while PECA 2016 defines the procedural standards and documentation required to support that recognition. Case law confirms that courts consistently reject evidence not because it is false, but because its authenticity, source, or completeness could not be demonstrated.

Based on this mapping, we should guide our investigation the following way.

- Maintain uninterrupted chain of custody documentation at every handover point, with no unexplained gaps.
- Always calculate and record hash values before and after imaging, and preserve them for court verification.

- Perform all analysis exclusively on verified image copies, never on the original device.
- Clearly document and disclose the source of all evidence at the point of collection.
- Submit complete evidence sets (e.g., full communication records, not partial logs) rather than fragments.

Ultimately, legal admissibility should be treated as a built-in outcome of correct technical procedure, not a separate concern addressed after the fact. A forensically sound process, properly documented, is what allows digital evidence to withstand legal scrutiny and serve its intended purpose in the pursuit of justice.